



Scenario YAML Files — SC-1 ... SC-8

One file per scenario in `scenarios/`. All use the Ch. 7 §7.3 DSL: `timeline` = deterministic script, `rules` = reactive triggers, one seeded RNG. `entity_groups` is the flagged loader extension (expands to N entities with deterministic ids `<prefix>_01..N` and position offsets). SC-2's seed **8841** is the golden-run fixture — do not change it after the Ch. 11 hash is recorded.

scenarios/sc1_night_patrol.yaml

```
name: "SC-1 Night Patrol Baseline"
area: "geo/sector_north.geojson"
sim_start: "2026-08-01T22:00:00Z"
duration_minutes: 120
seed: 42
difficulty: intro

entities:
  - { id: patrol_1, type: ground_team, callsign: "Nomad 1",
    start: [77.591, 12.972], behavior: patrol, patrol_zone: sec
    tor_a, personnel: 6 }
  - { id: patrol_2, type: ground_team, callsign: "Nomad 2",
    start: [77.605, 12.981], behavior: patrol, patrol_zone: sec
    tor_b, personnel: 6 }
  - { id: uav_1, type: drone, callsign: "Hawk 1", start:
    [77.598, 12.976], battery_pct: 100, behavior: idle }
  - { id: uav_2, type: drone, callsign: "Hawk 2", start:
    [77.600, 12.970], battery_pct: 100, behavior: idle }
  - { id: depot_n, type: depot, start: [77.588, 12.990], r
    esources: { fuel: 800, medkit: 20, rations: 200, battery: 1
    2 } }
entity_groups:
```

```

- { prefix: sensor, type: sensor_post, count: 7, along:
"geo/sensor_line_n.geojson", behavior: idle }

timeline:
- at: "+00:30"
  action: create_incident
  params: { kind: obstruction, severity: low, title: "Fallen tree on patrol track", near: patrol_1 }
- at: "+01:10"
  action: create_incident
  params: { kind: medical, severity: medium, title: "Sprained ankle – Nomad 2", near: patrol_2 }
- at: "+01:15"
  action: unit_report
  params: { unit: patrol_2, text: "One personnel injured during zone sweep, requesting guidance" }

rules:
- when: "random_chance(0.001)"
  then: { action: unit_report, params: { unit: patrol_1, text: "Sector quiet, nothing to report" } }

end_conditions:
- all_incidents_resolved: true
- sim_time_elapsed: true

```

scenarios/sc2_border_surveillance.yaml — ★ flagship & golden-run fixture

```

name: "SC-2 Border Surveillance & Infiltration Response"
area: "geo/border_sectors_ad.geojson"
sim_start: "2026-08-02T02:00:00Z"
duration_minutes: 360
seed: 8841 # FROZEN – golden-run fixture (Ch. 1)
difficulty: standard

entities:

```

```

- { id: cmd_post, type: command_post, callsign: "Ancho
r", start: [76.900, 32.100] }
- { id: qrf_1, type: ground_team, callsign: "Sabre
1", start: [76.905, 32.095], behavior: idle, personnel: 8 }
- { id: qrf_2, type: ground_team, callsign: "Sabre
2", start: [76.912, 32.093], behavior: idle, personnel: 8 }
- { id: uav_1, type: drone, callsign: "Hawk 1", star
t: [76.908, 32.098], battery_pct: 100, behavior: idle }
- { id: uav_2, type: drone, callsign: "Hawk 2", star
t: [76.915, 32.101], battery_pct: 100, behavior: idle }
- { id: depot_fwd, type: depot, start: [76.902, 32.097],
resources: { fuel: 1500, medkit: 35, rations: 400, battery:
24 } }
entity_groups:
- { prefix: patrol, type: ground_team, count: 4, zones:
[sector_a, sector_b, sector_c, sector_d], behavior: patrol,
personnel: 6 }
- { prefix: sensor, type: sensor_post, count: 12, along:
"geo/border_fence_line.geojson", behavior: idle }
- { prefix: op, type: observation_post, count: 8, alo
ng: "geo/ridge_line.geojson", behavior: idle, personnel: 3
}

timeline:
- at: "+00:20"
  action: unit_report
  params: { unit: sensor_04, text: "Seismic trip – sector
B, grid QB-4451" }
- at: "+00:22"
  action: unit_report
  params: { unit: sensor_05, text: "Second trip – adjacen
t segment, movement pattern consistent with 3-4 pax" }
- at: "+00:35"
  action: create_incident
  params: { kind: infiltration_suspected, severity: high,
title: "Suspected crossing – sector B fence line", near: se
nsor_04 }
- at: "+01:10"

```

```

    action: create_incident
    params: { kind: contact, severity: critical, title: "OP
-3 reports movement 400m north of post", near: op_03 }
  - at: "+02:00"
    action: spawn_weather
    params: { kind: fog, intensity: 0.6, center: [76.910, 3
2.102], radius_km: 4, drift_kmh: [2, -1] }
  - at: "+03:00"
    action: unit_report
    params: { unit: sensor_11, text: "Trip at sector D – po
ssible diversion attempt" }
  - at: "+03:05"
    action: create_incident
    params: { kind: infiltration_suspected, severity: high,
title: "Second crossing indication – sector D", near: senso
r_11 }

rules:
  - when: "unit_in_weather(uav_1, fog, 0.5)"
    then: { action: modify_speed, params: { unit: uav_1, mu
ltiplier: 0.5 } }
  - when: "resource_below(depot_fwd, fuel, 0.25)"
    then: { action: raise_alert, params: { severity: warnin
g, title: "Forward depot fuel below 25%" } }
  - when: "random_chance(0.0015)"
    then: { action: unit_report, params: { unit: op_05, tex
t: "Routine scan complete, negative contact" } }

end_conditions:
  - all_incidents_resolved: true
  - sim_time_elapsed: true

```

scenarios/sc3_counter_uav.yaml

```

name: "SC-3 Counter-UAV Airspace Watch"
area: "geo/airbase_perimeter.geojson"
sim_start: "2026-08-03T11:00:00Z"
duration_minutes: 240

```

```

seed: 3107
difficulty: standard

entities:
  - { id: cmd_post, type: command_post, callsign: "Vantage", start: [75.400, 30.200] }
  - { id: intercept_1, type: drone, callsign: "Falcon 1", start: [75.402, 30.198], battery_pct: 100, behavior: idle }
  - { id: intercept_2, type: drone, callsign: "Falcon 2", start: [75.405, 30.203], battery_pct: 100, behavior: idle }
  - { id: depot_ab, type: depot, start: [75.398, 30.201], resources: { fuel: 900, battery: 40, medkit: 10, rations: 150 } }
entity_groups:
  - { prefix: radar, type: sensor_post, count: 6, along: "geo/radar_ring.geojson", behavior: idle }
  - { prefix: guard, type: ground_team, count: 4, zones: [gate_n, gate_s, apron_e, apron_w], behavior: patrol, personnel: 4 }
  # intruder tracks are pre-authored routes – the engine flies them; CV pipeline classifies
  - { prefix: bogey, type: unidentified_uav, count: 3, routes: "geo/incursion_tracks.geojson", behavior: follow_route, spawn_at: ["+00:40", "+01:50", "+02:45"] }

timeline:
  - at: "+00:41"
    action: create_incident
    params: { kind: airspace_incursion, severity: high, title: "Unidentified UAV – restricted zone R-1", near: bogey_01 }
  - at: "+01:52"
    action: create_incident
    params: { kind: airspace_incursion, severity: high, title: "Second incursion – low-level approach from NE", near: bogey_02 }
  - at: "+02:46"
    action: create_incident

```

```

    params: { kind: airspace_incursion, severity: critical,
title: "Incursion over apron – vision review requested", ne
ar: bogey_03 }

rules:
  - when: "random_chance(0.001)"
    then: { action: unit_report, params: { unit: radar_02,
text: "Sweep clean, no tracks" } }

end_conditions:
  - all_incidents_resolved: true
  - sim_time_elapsed: true

```

scenarios/sc4_convoy_escort.yaml

```

name: "SC-4 Convoy Escort Under Threat"
area: "geo/valley_route_corridor.geojson"
sim_start: "2026-08-04T05:00:00Z"
duration_minutes: 300
seed: 5522
difficulty: standard

entities:
  - { id: convoy_lead, type: vehicle, callsign: "Packhorse
1", start: [77.100, 31.000], behavior: follow_route, route:
"geo/supply_route_main.geojson" }
  - { id: escort_1, type: ground_team, callsign: "Shield
1", start: [77.101, 31.001], behavior: follow_route, route:
"geo/supply_route_main.geojson", personnel: 6 }
  - { id: escort_2, type: ground_team, callsign: "Shield
2", start: [77.099, 30.999], behavior: follow_route, route:
"geo/supply_route_main.geojson", personnel: 6 }
  - { id: uav_1, type: drone, callsign: "Hawk 1", start: [7
7.102, 31.002], battery_pct: 100, behavior: follow_route, r
oute: "geo/route_overwatch.geojson" }
  - { id: depot_dest, type: depot, start: [77.350, 31.180],
resources: { fuel: 200, medkit: 8, rations: 60, battery: 6
} }

```

```

entity_groups:
  - { prefix: truck, type: vehicle, count: 6, follow: convo
y_lead, spacing_m: 80 }

timeline:
  - at: "+00:45"
    action: spawn_weather
    params: { kind: storm, intensity: 0.8, center: [77.220,
31.080], radius_km: 6, drift_kmh: [8, 3] }
  - at: "+01:30"
    action: create_incident
    params: { kind: threat_report, severity: high, title:
"Ambush risk assessed on pass segment K-7", near: convoy_le
ad }
  - at: "+02:10"
    action: create_incident
    params: { kind: obstruction, severity: medium, title:
"Rockfall blocking alternate track", near: truck_03 }

rules:
  - when: "unit_in_weather(convoy_lead, storm, 0.6)"
    then: { action: modify_speed, params: { unit: convoy_le
ad, multiplier: 0.5 } }
  - when: "unit_in_weather(uav_1, storm, 0.6)"
    then: { action: modify_speed, params: { unit: uav_1, mu
ltiplier: 0.4 } }

end_conditions:
  - unit_reached: { unit: convoy_lead, target: depot_dest }
  - sim_time_elapsed: true

```

scenarios/sc5_casualty_evacuation.yaml

```

name: "SC-5 Forward Post Casualty Evacuation"
area: "geo/forward_posts_two.geojson"
sim_start: "2026-08-05T04:30:00Z"
duration_minutes: 180
seed: 9014

```

```

difficulty: hard

entities:
  - { id: post_a, type: command_post, callsign: "Bastion A", start: [76.500, 33.000] }
  - { id: post_b, type: command_post, callsign: "Bastion B", start: [76.560, 33.040] }
  - { id: medevac_1, type: vehicle, callsign: "Angel 1", start: [76.520, 33.010], behavior: idle }
  - { id: medevac_2, type: vehicle, callsign: "Angel 2", start: [76.522, 33.012], behavior: idle }
  - { id: fld_hospital, type: depot, start: [76.518, 33.008], resources: { medkit: 30, fuel: 400, rations: 100, battery: 8 } }
entity_groups:
  - { prefix: sect_a, type: ground_team, count: 5, zones: [post_a_perimeter], behavior: patrol, personnel: 6 }
  - { prefix: sect_b, type: ground_team, count: 5, zones: [post_b_perimeter], behavior: patrol, personnel: 6 }

timeline:
  - at: "+00:15"
    action: create_incident
    params: { kind: contact, severity: critical, title: "Engagement at Bastion A perimeter", near: post_a }
  - at: "+00:25"
    action: create_incident
    params: { kind: medical, severity: critical, title: "2 casualties – Bastion A, medevac requested", near: post_a }
  - at: "+00:55"
    action: create_incident
    params: { kind: medical, severity: critical, title: "3 casualties – Bastion B, simultaneous demand", near: post_b }
  - at: "+01:20"
    action: create_incident
    params: { kind: medical, severity: high, title: "Walking wounded – Bastion B secondary", near: post_b }

```

```

rules:
  - when: "resource_below(fld_hospital, medkit, 0.3)"
    then: { action: raise_alert, params: { severity: critical, title: "Field hospital medkits below 30% – saturation imminent" } }

end_conditions:
  - all_incidents_resolved: true
  - sim_time_elapsed: true

```

scenarios/sc6_logistics_sustainment.yaml

```

name: "SC-6 Extended Logistics Sustainment"
area: "geo/forward_line_corridor.geojson"
sim_start: "2026-08-06T00:00:00Z"
duration_minutes: 720
seed: 6633
difficulty: standard

entities:
  - { id: depot_main, type: depot, start: [76.000, 32.500],
    resources: { fuel: 5000, medkit: 120, rations: 2000, battery: 80 } }
  - { id: resupply_1, type: vehicle, callsign: "Mule 1", start: [76.001, 32.501], behavior: idle }
  - { id: resupply_2, type: vehicle, callsign: "Mule 2", start: [76.002, 32.499], behavior: idle }
entity_groups:
  - { prefix: fwd_post, type: depot, count: 5, along: "geo/forward_posts_line.geojson",
    resources: { fuel: 300, medkit: 10, rations: 120, battery: 6 } }
  - { prefix: patrol, type: ground_team, count: 10, zones: [fwd_zones_all], behavior: patrol, personnel: 6 }
  - { prefix: uav, type: drone, count: 4, start_near: depot_main, battery_pct: 100, behavior: idle }

```

```

timeline:
  - at: "+04:00"
    action: create_incident
    params: { kind: threat_report, severity: medium, title:
"Route reassessment doubles patrol tempo – sectors 2-4" }
    # consumption pressure comes from the logistics system it
self; this scenario is deliberately script-light

rules:
  - when: "resource_below(fwd_post_02, fuel, 0.2)"
    then: { action: raise_alert, params: { severity: warnin
g, title: "FP-2 fuel below 20%" } }
  - when: "resource_below(fwd_post_04, medkit, 0.25)"
    then: { action: raise_alert, params: { severity: warnin
g, title: "FP-4 medical stock below 25%" } }
  - when: "random_chance(0.0008)"
    then: { action: create_incident, params: { kind: vehicl
e_breakdown, severity: medium, title: "Resupply vehicle bre
akdown" } }

end_conditions:
  - sim_time_elapsed: true

```

scenarios/sc7_multi_sector_escalation.yaml

```

name: "SC-7 Multi-Sector Escalation Exercise"
area: "geo/border_sectors_ad.geojson"
sim_start: "2026-08-07T18:00:00Z"
duration_minutes: 480
seed: 7788
difficulty: hard          # war-room exercise – designed for
Ch. 15 sessions + Ch. 16 RTA

entities:
  - { id: cmd_post, type: command_post, callsign: "Anchor",
start: [76.900, 32.100] }
  - { id: qrf_1, type: ground_team, callsign: "Sabre 1", st
art: [76.905, 32.095], behavior: idle, personnel: 8 }

```

```

- { id: qrf_2, type: ground_team, callsign: "Sabre 2", start: [76.912, 32.093], behavior: idle, personnel: 8 }
- { id: depot_fwd, type: depot, start: [76.902, 32.097], resources: { fuel: 2000, medkit: 50, rations: 600, battery: 30 } }
entity_groups:
- { prefix: patrol, type: ground_team, count: 12, zones: [sector_a, sector_b, sector_c, sector_d], behavior: patrol, personnel: 6 }
- { prefix: sensor, type: sensor_post, count: 24, along: "geo/border_fence_line.geojson", behavior: idle }
- { prefix: op, type: observation_post, count: 12, along: "geo/ridge_line.geojson", behavior: idle, personnel: 3 }
- { prefix: uav, type: drone, count: 6, start_near: cmd_post, battery_pct: 100, behavior: idle }

timeline:
  # deliberate unit shortage: 4 sectors light up, QRF can cover 2
  - at: "+00:30"
    action: create_incident
    params: { kind: infiltration_suspected, severity: high, title: "Sector A fence trip cluster", near: sensor_03 }
  - at: "+00:42"
    action: create_incident
    params: { kind: contact, severity: critical, title: "Sector C OP contact report", near: op_07 }
  - at: "+00:55"
    action: create_incident
    params: { kind: infiltration_suspected, severity: high, title: "Sector D trip – pattern ambiguous", near: sensor_21 }
  - at: "+01:05"
    action: create_incident
    params: { kind: medical, severity: critical, title: "Casualty during Sector C response", near: op_07 }
  - at: "+01:30"
    action: spawn_weather

```

```

    params: { kind: storm, intensity: 0.7, center: [76.915,
32.105], radius_km: 8, drift_kmh: [6, -2] }
    - at: "+02:15"
      action: create_incident
      params: { kind: infiltration_suspected, severity: high,
title: "Sector B secondary crossing indication", near: sens
or_09 }

rules:
  - when: "resource_below(depot_fwd, fuel, 0.3)"
    then: { action: raise_alert, params: { severity: warnin
g, title: "Forward depot fuel below 30%" } }
  - when: "random_chance(0.002)"
    then: { action: unit_report, params: { unit: op_02, tex
t: "Movement unconfirmed, observing" } }

end_conditions:
  - sim_time_elapsed: true # graded exercise: score come
s from the after-action report, not a win state

```

scenarios/sc8_hadr_flood_relief.yaml

```

name: "SC-8 HADR Deployment – Flood Relief (aid to civil au
thority)"
area: "geo/river_district.geojson"
sim_start: "2026-08-08T06:00:00Z"
duration_minutes: 360
seed: 2450
difficulty: standard
# Retained per Ch. 19 §19.1: HADR is a standing military mi
ssion and the only
# scenario class with open real-world outcome data (Ch. 14
benchmarking link).

entities:
  - { id: cmd_post, type: command_post, callsign: "Relief A
ctual", start: [76.340, 10.010] }
  - { id: eng_1, type: ground_team, callsign: "Sapper 1", s

```

```

tart: [76.342, 10.012], behavior: idle, personnel: 10 }
  - { id: medevac_1, type: vehicle, callsign: "Angel 1", start: [76.341, 10.009], behavior: idle }
  - { id: depot_stage, type: depot, start: [76.338, 10.011], resources: { fuel: 1200, medkit: 60, rations: 1500, battery: 20 } }
entity_groups:
  - { prefix: evac_team, type: ground_team, count: 6, zones: [district_zones], behavior: idle, personnel: 8 }
  - { prefix: boat, type: vehicle, count: 4, start_near: cmd_post, behavior: idle }
  - { prefix: uav, type: drone, count: 3, start_near: cmd_post, battery_pct: 100, behavior: idle }

timeline:
  - at: "+00:10"
    action: spawn_weather
    params: { kind: rain, intensity: 0.9, center: [76.345, 10.015], radius_km: 10, drift_kmh: [3, 1] }
  - at: "+00:40"
    action: create_incident
    params: { kind: evacuation_request, severity: critical, title: "Ward 4 cut off – evacuation tasking from civil authority", near: evac_team_02 }
  - at: "+01:20"
    action: create_incident
    params: { kind: medical, severity: high, title: "Medical camp overload – Ward 7", near: medevac_1 }
  - at: "+02:00"
    action: create_incident
    params: { kind: obstruction, severity: high, title: "Bridge unsafe – engineering assessment required", near: eng_1 }

rules:
  - when: "unit_in_weather(uav_1, rain, 0.7)"
    then: { action: modify_speed, params: { unit: uav_1, multiplier: 0.6 } }

```

```
- when: "resource_below(depot_stage, rations, 0.25)"
  then: { action: raise_alert, params: { severity: warning, title: "Staging depot rations below 25%" } }

end_conditions:
- all_incidents_resolved: true
- sim_time_elapsed: true
```

Loader notes (Phase 1 tasks these files create)

- ☐ Implement `entity_groups` expansion (deterministic ids + placement along GeoJSON lines / zones / follow-spacing)
- ☐ Add DSL actions used above beyond the Ch. 7 base set to the whitelist: `unit_reached` end-condition, `spawn_at` for pre-authored routes — both trivial, both must be whitelisted explicitly
- ☐ Author the referenced `geo/*.geojson` files (draw in geojson.io — free, ~1 hour total)
- ☐ Record SC-2 golden-run hash (seed 8841, 100 ticks) into the Ch. 11 test suite